

SQL · DATA ANALYSIS PORTFOLIO



PLATE & LEDGER

Order Analytics



An end-to-end SQL exploration of a restaurant's menu & order data —
uncovering pricing patterns, ordering behavior, and customer spend
through structured query analysis across three objectives.

DATABASE

restaurant_db

ENGINE

MySQL 8.0

TABLES

menu_items · order_details

Menu Overview

The `menu_items` table holds every dish served by the restaurant, along with its category and price. We begin by profiling the table's size and its pricing extremes.

32

TOTAL MENU ITEMS

4

MENU CATEGORIES

\$5.00 – \$19.95

PRICE RANGE

1 How many items are on the menu?

SQL

```
-- 1. Find the number of items on the menu
SELECT COUNT(item_name) FROM menu_items;
```

RESULT

COUNT(item_name)
32

2 What are the least and most expensive items on the menu?

SQL

```
-- 2. Least and most expensive items on the menu
SELECT item_name AS least_price_item, price FROM menu_items
ORDER BY price ASC LIMIT 1;
SELECT item_name AS most_exp_item, price FROM menu_items
ORDER BY price DESC LIMIT 1;
```

LEAST EXPENSIVE

Item	Price
Edamame	\$5.00

MOST EXPENSIVE

Item	Price
Shrimp Scampi	\$19.95

Insight — The menu spans a **\$14.95 price range**, from the plant-based Edamame appetizer to the premium Shrimp Scampi entrée, suggesting a deliberate mix of budget-friendly starters and higher-margin signature dishes.

Spotlight: Italian Cuisine

Narrowing the focus to the **Italian** category to understand its size and price positioning relative to the rest of the menu.

3 How many Italian dishes are on the menu?

SQL

```
-- 3. How many Italian dishes are on the menu?
SELECT COUNT(item_name) AS italian_dishes FROM menu_items
WHERE category = 'Italian';
```

RESULT

italian_dishes
9

4 What are the least and most expensive Italian dishes?

SQL

```
-- 4. Least and most expensive Italian dishes
SELECT item_name AS least_italian_dish, price FROM menu_items
WHERE category = 'Italian'
ORDER BY price ASC LIMIT 2;

SELECT item_name AS exp_italian_dish, price FROM menu_items
WHERE category = 'Italian'
ORDER BY price DESC LIMIT 1;
```

LEAST EXPENSIVE (TIE)

Item	Price
Spaghetti	\$14.50
Fettuccine Alfredo	\$14.50

MOST EXPENSIVE

Item	Price
Shrimp Scampi	\$19.95

Insight — Italian is tied for the **largest category** on the menu (9 dishes) and also contains the single most expensive item on the entire menu — Shrimp Scampi doubles as both the priciest Italian dish and the priciest dish overall.

Category Breakdown

Grouping the full menu by category reveals how dish count and average price vary across American, Asian, Mexican, and Italian offerings.

5 How many dishes are in each category?

SQL

```
-- 5. How many dishes are in each category?
SELECT category, COUNT(item_name) FROM menu_items
GROUP BY category;
```



6 What is the average dish price within each category?

SQL

```
-- 6. Average dish price within each category
SELECT category, ROUND(AVG(price),2) AS avg_price FROM menu_items
GROUP BY category;
```

RESULT — SORTED BY AVERAGE PRICE

Category	Dishes	Avg. Price
Italian	9	\$16.75
Asian	8	\$13.48
Mexican	9	\$11.80
American	6	\$10.07

Insight — Italian dishes carry the **highest average price** (\$16.75) despite tying for the most dishes on the menu, while American — the smallest category — is also the most budget-friendly at \$10.07 on average. This points to Italian as the restaurant's premium category.

Order Volume & Timeframe

The **order_details** table logs every item sold, tied to an order_id, date, and time. We first establish the window of activity and the overall order volume.

Jan 1 – Mar 31,
2023

DATE RANGE (90 DAYS)

5,370

TOTAL ORDERS

12,234

TOTAL ITEMS ORDERED

1 View the order_details table — what is its date range?

SQL

```
-- 1. View the order_details table
SELECT * FROM order_details;

-- 2. What is the date range of the table?
SELECT MIN(order_date) AS minimum_date,
       MAX(order_date) AS maximum_date
FROM order_details;
```

RESULT

minimum_date	maximum_date
2023-01-01	2023-03-31

2 How many orders — and how many items — fall within this date range?

SQL

```
-- 3. How many orders were made within this date range?
SELECT COUNT(DISTINCT(order_id)) AS num_orders FROM order_details;

-- 4. How many items were ordered within this date range?
SELECT COUNT(*) AS total_items FROM order_details;
```

DISTINCT ORDERS

num_orders
5,370

ITEMS ORDERED

total_items
12,234

Insight — Over the 90-day window the restaurant logged roughly **60 orders per day** and **2.3 items per order** on average ($12,234 \div 5,370$) — a picture of frequent, small-basket ordering typical of a casual dining venue.

Largest Orders

Grouping by order_id surfaces the busiest single orders — and how rare genuinely large orders are across the full 5,370-order dataset.

3 Which orders had the most number of items?

SQL

```
-- 5. Which orders had the most number of items?  
SELECT order_id, COUNT(item_id) AS most_ordered  
FROM order_details  
GROUP BY order_id  
ORDER BY most_ordered DESC;
```

RESULT — TOP ORDERS BY ITEM COUNT

order_id	most_ordered
330	14
440	14
443	14
1957	14
2675	14
3473	14
405	14
1274	13

4 How many orders had more than 12 items?

SQL

```
-- 6. How many orders had more than 12 items?  
SELECT COUNT(*) FROM (  
  SELECT order_id, COUNT(item_id) AS most_ordered  
  FROM order_details  
  GROUP BY order_id  
  HAVING most_ordered > 12  
) AS total_orders;
```

RESULT

total_orders
20

Insight — Just **20 of 5,370 orders (0.37%)** exceeded 12 items, and seven orders are tied at the observed maximum of 14 — large group orders are a rare but consistent occurrence rather than one-off outliers.

Joining Menu & Orders

Joining `order_details` to `menu_items` attaches item name, category, and price to every line ordered — the foundation for understanding what customers actually buy.

1 Combine the `menu_items` and `order_details` tables into a single table.

SQL

```
-- 1. Combine menu_items and order_details into a single table
SELECT * FROM order_details
LEFT JOIN menu_items
  ON order_details.item_id = menu_items.menu_item_id;
```

RESULT — SAMPLE OF 12,234 COMBINED ROWS

order_id	order_date	item_name	category	price
1	2023-01-01	Korean Beef Bowl	Asian	\$17.95
2	2023-01-01	Tofu Pad Thai	Asian	\$14.50
2	2023-01-01	Spaghetti	Italian	\$14.50
2	2023-01-01	Chicken Burrito	Mexican	\$12.95
2	2023-01-01	Mushroom Ravioli	Italian	\$15.50

2 What were the least and most ordered items — and their categories?

SQL

```
-- 2. Least and most ordered items, with category
SELECT item_name, category, COUNT(order_details_id) AS least_purchased
FROM order_details LEFT JOIN menu_items ON order_details.item_id = menu_items.menu_item_id
GROUP BY item_name, category
ORDER BY least_purchased ASC LIMIT 1;

SELECT item_name, category, COUNT(order_details_id) AS most_purchased
FROM order_details LEFT JOIN menu_items ON order_details.item_id = menu_items.menu_item_id
GROUP BY item_name, category
ORDER BY most_purchased DESC LIMIT 1;
```

LEAST ORDERED

Item	Category	Orders
Chicken Tacos	Mexican	123

MOST ORDERED

Item	Category	Orders
Hamburger	American	622

Insight — The Hamburger — the cheapest American dish — was ordered **5x more often** than the least-popular Chicken Tacos (622 vs. 123), showing customers strongly favor familiar, budget-friendly staples over niche menu items.

Top Spending Orders

Summing price across each order_id ranks orders by total dollars spent, independent of item count.

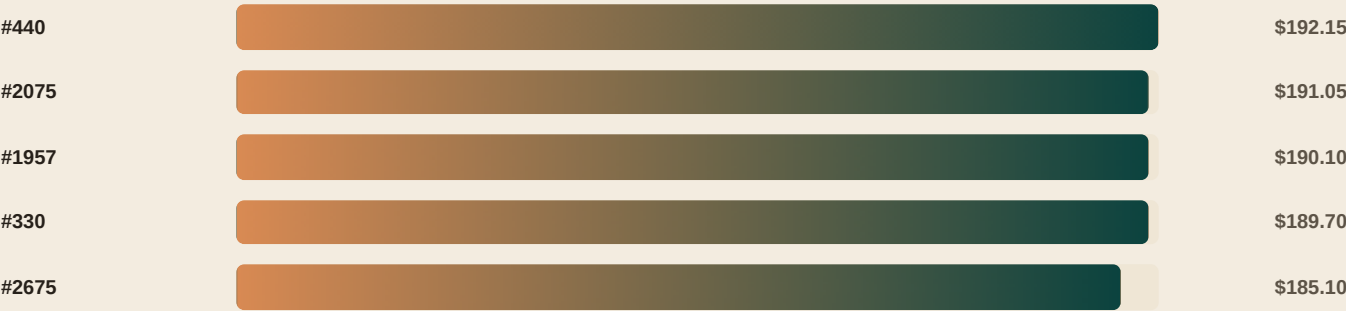
3 What were the top 5 orders that spent the most money?

SQL

```
-- 3. Top 5 orders by total spend
SELECT order_id, SUM(price) AS total_spent
FROM order_details LEFT JOIN menu_items ON order_details.item_id = menu_items.menu_item_id
GROUP BY order_id
ORDER BY total_spent DESC
LIMIT 5;
```

RESULT

order_id	total_spent
440	\$192.15
2075	\$191.05
1957	\$190.10
330	\$189.70
2675	\$185.10



Insight — The top 5 orders sit within a tight **\$7 band** (\$185–\$192), and every one of them also appears among the seven 14-item orders from Objective 2 — the biggest baskets and the biggest bills are, unsurprisingly, the same orders.

Inside the Highest Spenders

Breaking the top orders down by category shows *why* they cost so much — not just how many items they contained.

4 What's inside the single highest-spend order (#440)?

SQL

```
-- 4. Detail of the highest-spend order
SELECT category, COUNT(item_id) AS num_items
FROM order_details LEFT JOIN menu_items ON order_details.item_id = menu_items.menu_item_id
WHERE order_id = 440
GROUP BY category;
```

RESULT — ORDER #440 (14 ITEMS · \$192.15)

Category	Items
Italian	8
Mexican	2
American	2
Asian	2

5 Bonus — same breakdown across the top 5 highest-spend orders

SQL

```
-- 5. Category detail across the top 5 highest-spend orders
SELECT order_id, category, COUNT(item_id) AS num_items
FROM order_details LEFT JOIN menu_items ON order_details.item_id = menu_items.menu_item_id
WHERE order_id IN (440, 2075, 1957, 330, 2675)
GROUP BY order_id, category;
```

RESULT — CATEGORY MIX BY ORDER

order_id	Category	Items
440	Italian	8
440	Mexican / American / Asian	2 each
1957	Italian	5
1957	Asian / American / Mexican	3 each
330	Asian	6
330	Mexican	4
330	Italian	3
330	American	1

Insight — In 2 of the 3 highest-spend orders with visible detail, **Italian dishes make up the single largest share of items** (8 of 14 in #440; 5 of 14 in #1957) — reinforcing the Objective 1 finding that Italian is the menu's premium category and the clearest driver of high-value orders.

KEY TAKEAWAYS

Menu, By the Numbers

Across three objectives — the menu, 90 days of orders, and combined customer behavior — the data tells one consistent story: a menu with an accessible American core and a premium Italian tier, and a customer base that mostly keeps it simple but occasionally orders big, Italian-heavy, high-spend baskets.

- Menu: 32 items across 4 categories, priced \$5.00 – \$19.95
- Italian is the priciest category (\$16.75 avg) and ties for the most dishes (9)
- Orders: 5,370 orders / 12,234 items placed between Jan 1 – Mar 31, 2023
- Only 20 orders (0.37%) topped 12 items; 7 orders tie at a max of 14
- Hamburger is the most-ordered item (622x); Chicken Tacos the least (123x)
- The top 5 highest-spend orders (\$185–\$192) are also the largest by item count, and lean heavily on Italian dishes

Queries written & executed in MySQL Workbench 8.0 | restaurant_db.menu_items